TaskFlow Pro — MVP SRS

Project Name	TaskFlow Pro
Document Type	Minimum Viable Product (MVP) Software Requirements Specification
Technology Stack	MERN Stack (MongoDB, Express.js, React.js, Node.js) + Tailwind CSS
Version / Target	v1.0 (Condensed Baseline for Feature Expansion)

1. MVP Project Overview & Objectives

TaskFlow Pro is an enterprise workforce task management and performance monitoring platform tailored for IT departments. The Minimum Viable Product (MVP) version focuses on eliminating fragmented manual processes (such as unmanaged spreadsheets and disjointed emails) by establishing a structured, real-time core system for core project allocation and oversight.

Core Objectives

- **Centralize Task Management:** Consolidate core assignment workflows into a uniform interface.
- **Establish Accountability:** Provide absolute transparent verification over task ownership and workflow milestones.
- **Real-Time Visibility:** Deliver immediate insight into current project deployment progress states.

2. Core Actors & Permissions

- **Administrator:** Possesses systemic management privileges. Key permissions include managing employee profiles, orchestrating core task life cycles (creation, assignment, updates, deletion), viewing live organizational progress metrics, and reviewing submitted technical proofs.
- **IT Employee:** Possesses operational execution privileges restricted strictly to personal work scopes. Key permissions include viewing individual task boards, providing progress incremental modifications, uploading verification documentation, and engaging via task threads.

3. MVP Functional Requirements

3.1 Authentication & Session Management

- Secure login and logout infrastructure leveraging JSON Web Tokens (JWT) for stateless identity verification.
- Password hashing implemented at the database tier using bcrypt.

- Basic operations for administrators to securely provision, update, search, or systematically deactivate employee user profiles.

3.2 Task, Progress & Status Engineering

- **Administrative Lifecycle Matrix:** Complete pipeline management enabling administrators to create, assign, update, reassign, or delete work records.
- **Task Tracking Increments:** Hardcoded percentage granular updates limited to distinct increments: `0%`, `25%`, `50%`, `75%`, and `100%`.
- **Standard Status States:** A consolidated tracking workflow comprising: `New`, `Assigned`, `In Progress`, `Under Review`, `Completed`, and `Overdue`.

3.3 Collaboration, File Handling & Automated Alerts

- **File Management Infrastructure:** Support for employee uploads of explicit technical validations (such as PDF metrics, logs, or diagnostic screenshots) reviewable by administrators.
- **Contextual Task Comments:** Direct communication threads attached directly to distinct task records to ensure clean audit trails between administrative managers and assigned technicians.
- **Automated Notifications:** Native internal alerts triggered directly by explicit data updates: task creation/assignment, upcoming deadlines, overdue transitions, or completion submissions.

4. Technical Architecture & Database Design

The system utilizes a structured modular architecture to support future expansions seamlessly. The presentation tier is built using React.js and Tailwind CSS; the application backend uses Express.js running on Node.js; data persistence is managed natively within MongoDB, coupled with a dedicated cloud engine for attachments.

Core Database Schema Layout

User Collection

```
- id
- fullName
- email
- password (hashed)
- role (Admin / Employee)
- department
- status (Active/Inactive)
- createdAt
```

Task Collection

```
- id
- title
- description
- assignedTo
- assignedBy
- status
- progress (0% - 100%)
- startDate
- dueDate
- attachments
```

Support Collections

```
Comment:
- id, taskId, userId,
message, createdAt

Notification:
- id, userId, title,
message, isRead, createdAt
```

5. Primary MVP API Specifications

Route Endpoint	HTTP Method	Purpose / Functional Mapping
`/api/auth/login`	POST	Authenticates credentials and issues signed JWT tokens.
`/api/auth/logout`	POST	Destroys active sessions and clears client context.
`/api/employees`	POST / GET	Enables administrators to provision or list employee profiles.
`/api/employees/{id}`	PUT / DELETE	Updates employee data points or triggers system deactivation.
`/api/tasks`	POST / GET	Creates new tasks or queries systemic task repositories.
`/api/tasks/{id}`	PUT / DELETE	Modifies core task criteria or removes a task record.
`/api/tasks/{id}/progress`	PATCH	Updates task completion progress percentages (0% - 100%).
`/api/tasks/{id}/status`	PATCH	Transitions a task between workflow operational status flags.
`/api/tasks/{id}/comments`	POST / GET	Appends comments or fetches conversation histories.
`/api/files/upload`	POST	Handles streaming uploads of screenshots or verification PDFs.
`/api/dashboard/admin`	GET	Retrieves top-level organizational KPIs for management.
`/api/dashboard/employee`	GET	Fetches specific personal task indicators for workers.

6. Deferred Post-MVP Roadmap

To preserve technical agility and guarantee speed of deployment, advanced enterprise metrics and integrations have been decoupled from the initial runtime build. These remain fully compatible with the base models and are planned for Phase 2 implementation:

- **Advanced Rules Engine:** Granular Priority matrices (Critical/High/Medium/Low), Multi-category tags (Network, Security, DevOps, etc.), and systemic Activity Logs.
- **Enterprise Governance:** Three-tiered Role-Based Access Control (Super Admin hierarchies) and interactive Leave & Employee Availability management.
- **Intelligent Enhancements & Ecosystems:** AI-driven task automated assignment, automated prediction modeling, native Slack/Teams integrations, and standalone mobile companion architectures.

This MVP document provides a clean, backward-compatible framework that mirrors the layout of your main technical documentation, allowing you to easily layer on advanced features later without breaking core patterns.